



▲ FAIRNESS IMPROVED

● NO ACCURACY LOSS

Executive Summary

Fairness Experiment Report - Upload 33 summarizes the current fairness state for upload 33. Accuracy changed to +0.838 and fairness score changed to +0.953. Strong fairness improvement with no measurable accuracy loss. The best balanced model is ThresholdOptimizer_mitigated_threshold with combined score 0.907.

ACCURACY	FAIRNESS SCORE	COMBINED SCORE	DISPARATE IMPACT
0.838 ▲ +0.838	0.953 ▲ +0.953	0.907 ThresholdOptimizer_mitigated_threshold	0.575 Ratio (DIR)

Dataset & Model Information

DATASET	MODEL	SECTIONS INCLUDED
processed_dataset.csv	log_reg_baseline.pkl	comparison

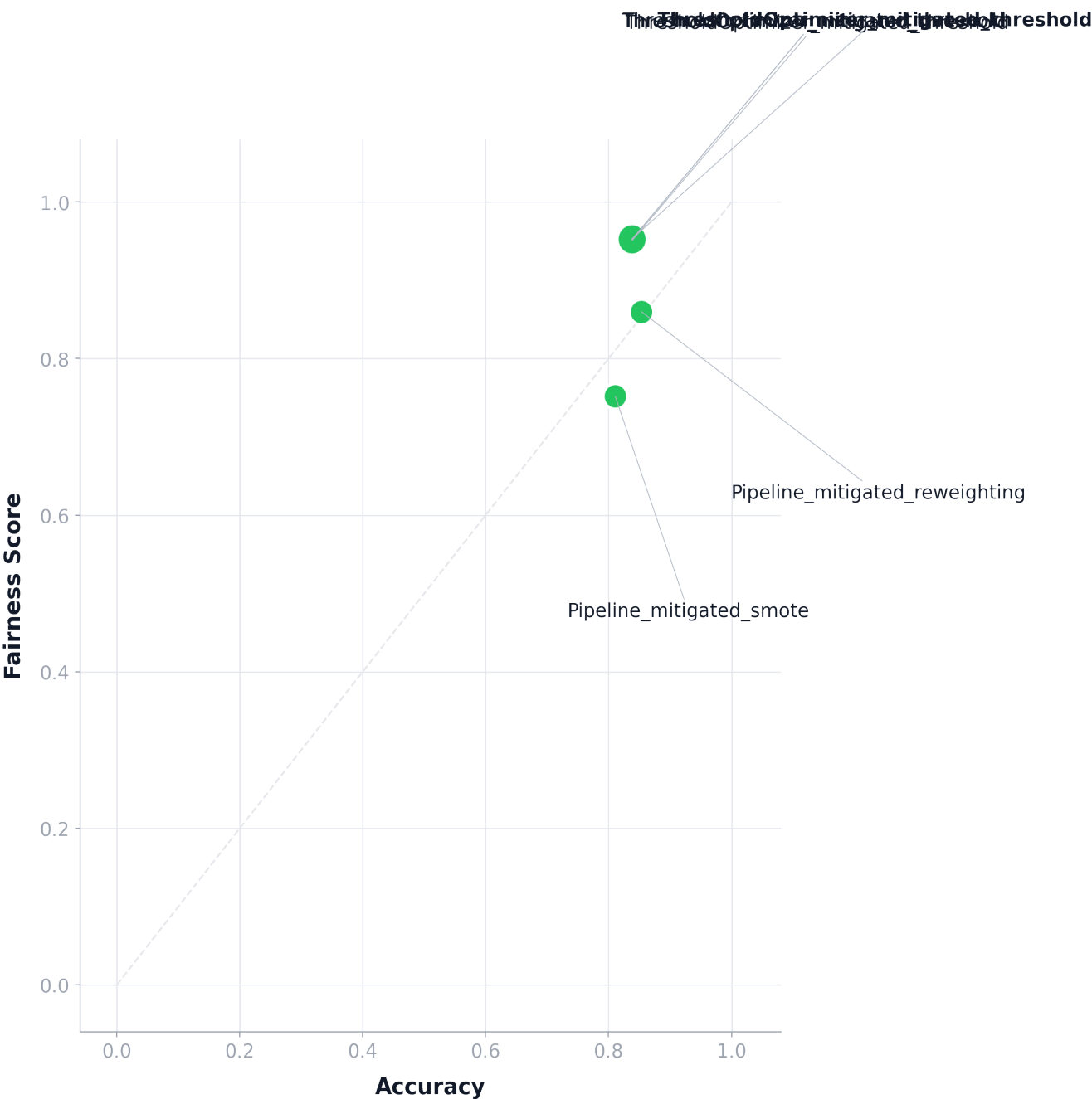
Fairness Metrics

Metric	Value	Interpretation
Demographic Parity Difference (DPD)	0.088	Difference in positive-prediction rates across groups; closer to 0 is more equitable.
Equalized Odds Difference (EOD)	0.006	Gap in true/false positive rates between groups; closer to 0 indicates equal error rates.
Disparate Impact Ratio (DIR)	0.575	Ratio of favorable outcome rates between groups; values near 1.0 indicate parity.

Model Comparison

Model	Accuracy	Fairness	DPD	EOD	DIR	Combined
★ ThresholdOptimizer_mitigated_threshold	0.838	0.953	0.088	0.006	0.575	0.907
★ ThresholdOptimizer_mitigated_threshold	0.838	0.952	0.088	0.007	0.574	0.907
★ ThresholdOptimizer_mitigated_threshold	0.838	0.952	0.087	0.008	0.576	0.906
★ ThresholdOptimizer_mitigated_threshold	0.839	0.951	0.090	0.008	0.566	0.906
Pipeline_mitigated_reweighting	0.853	0.860	0.180	0.101	0.296	0.857
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Pipeline_mitigated_smote	0.811	0.752	0.332	0.164	0.285	0.775
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Accuracy vs. Fairness Trade-off



Recommendations

Fairness improved after mitigation. Consider validating and deploying the mitigated model.